IBM University Engagement Project Submission

"Exploring the Power of Agentic AI with IBM Granite and LangFlow"

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Project Tile – AI Agent for Smart Farming Advice

Domain of Project – Agriculture

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Problem Statement - AI Agent for Smart Farming Advice

The Challenge

Farmers often struggle with low crop yield, unpredictable weather conditions, pest infestations, soil health issues, and limited access to timely agricultural expertise. Traditional farming advisory services are fragmented, slow, and not personalized to the farmer's specific crop, soil type, location, or climatic conditions. As a result, farmers may face reduced productivity, increased costs, and lower profitability.

With the advancement of Artificial Intelligence and Generative AI, there is a need for an intelligent farming assistant that can provide real-time, personalized, and data-driven recommendations to support better agricultural decision-making.

The Objective

Develop an **AI-Powered Smart Farming Advice Agent** using **Langflow** and **IBM watsonx.ai** that provides personalized, real-time, and sustainable agricultural guidance to farmers.

The proposed solution should:

1. **Agricultural Knowledge Retrieval (RAG)**
   - a. Retrieve crop cultivation practices, government advisories, and agricultural guidelines from a knowledge base.
   - b. Provide accurate answers to farming-related questions.
2. **Crop Advisory System**
   - a. Recommend suitable crops based on soil type, season, and environmental conditions.
   - b. Suggest seeds, fertilizers, and cultivation techniques.
3. **Weather and Irrigation Guidance**
   - a. Provide irrigation recommendations based on crop requirements.
   - b. Offer weather-based farming advice to improve productivity.
4. **Pest and Disease Advisory**
   - a. Assist farmers in identifying common crop pests and diseases.
   - b. Suggest preventive measures and treatment recommendations.

5. Market Insights

    a. Provide information about crop demand, pricing trends, and selling opportunities.

    b. Help farmers make informed marketing decisions.

6. Multi-Agent Architecture

    a. Crop Advisory Agent

    b. Weather & Irrigation Agent

    c. Pest & Disease Advisory Agent

    d. Market Insights Agent

Technology Used

LangFlow orchestrates the multi-agent workflow, IBM watsonx.ai provides the AI platform, and Granite Models deliver intelligent reasoning, emotion analysis, and personalized mental health support.

I used LangFlow + IBM watsonx.ai + Granite Models

LangFlow serves as the visual orchestration platform for designing and managing the multi-agent workflow. It enables seamless integration of AI components through a drag-and-drop interface, facilitating prompt engineering, workflow automation, agent coordination, and interaction management. LangFlow simplifies the development of complex Agentic AI applications without extensive coding.

IBM watsonx.ai provides the enterprise-grade AI development environment that powers the solution's intelligence layer. It offers access to foundation models, prompt engineering capabilities, model deployment, and governance features, enabling secure and scalable AI-driven mental health support.

IBM Granite Models act as the core Large Language Models (LLMs) responsible for understanding user inputs, analyzing emotions, assessing mental health risks, and generating context-aware responses. These models provide natural language understanding and reasoning capabilities essential for mental health awareness and suicide prevention.

Proposed Solution

AI Agent for Smart Farming Advice

The proposed solution is an **AI Agent for Smart Farming Advice** developed using **Langflow**, **IBM watsonx.ai**, and **IBM Granite Models**. The system acts as an intelligent virtual agricultural assistant that provides farmers with personalized recommendations and real-time guidance for effective farm management.

The AI agent leverages **Retrieval-Augmented Generation (RAG)** to access agricultural knowledge bases, farming guidelines, government advisories, and best practices. By analyzing farmer queries, crop information, soil conditions, and environmental factors, the system generates accurate and context-aware recommendations.

The proposed system consists of multiple specialized AI agents working collaboratively:

1. Crop Advisory Agent

- Recommends suitable crops based on soil type, season, and climatic conditions.
- Provides information on seed selection, sowing methods, and crop management practices.

2. Weather & Irrigation Agent

- Analyzes weather forecasts and crop water requirements.
- Suggests irrigation schedules and water conservation techniques.

3. Pest & Disease Advisory Agent

- Identifies common crop pests and diseases.
- Recommends preventive measures and suitable treatments.
- Supports image-based disease identification for future enhancements.

4. Market Insights Agent

- Provides information about market trends, crop demand, and pricing.
- Helps farmers make informed decisions regarding harvesting and selling produce.

5. Agricultural Knowledge Retrieval (RAG)

- Retrieves relevant information from agricultural documents, government advisories, and farming manuals.
- Generates accurate answers to farmer queries using IBM Granite Models.

Langflow component Used

For **Agentic AI for Mental Health Awareness & Suicide Prevention**, you can present the components in a more professional and impactful way:

1. User Interaction (Chat Input)

Serves as the primary interface where users share thoughts, emotions, concerns, or mental health-related queries through natural language conversations.

2. IBM watsonx.ai Agent

Acts as the intelligent decision-making engine, analyzing user inputs, assessing emotional states, identifying potential mental health risks, and generating personalized support strategies.

3. IBM Granite Model Granite-4.0-8B-Instruct

A powerful and efficient Large Language Model (LLM) that enables real-time emotional understanding, contextual reasoning, suicide risk assessment, and personalized mental health guidance.

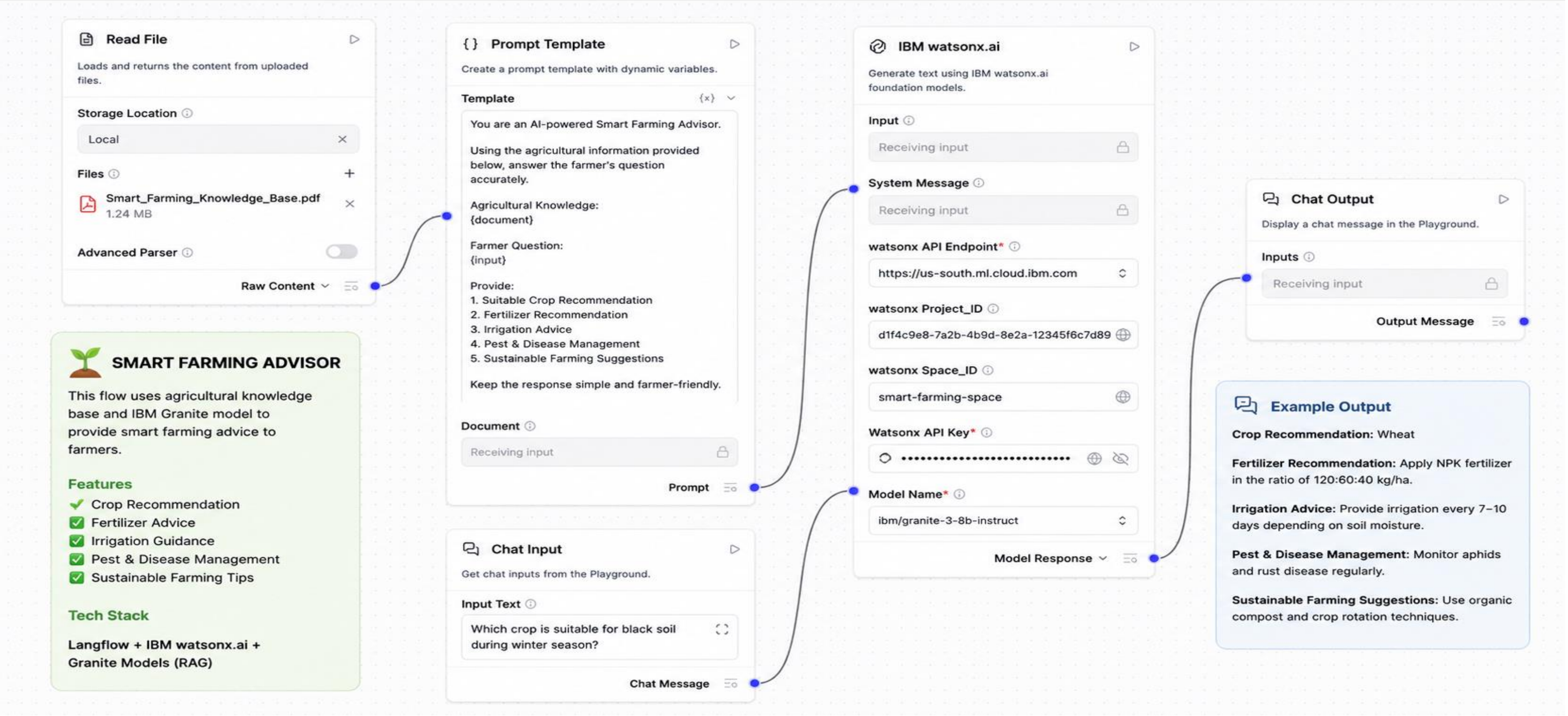
4. AI Response Interface (Chat Output)

Delivers empathetic, context-aware responses including mental health awareness content, coping strategies, wellness recommendations, emotional support, and crisis intervention guidance.

5. Knowledge & Data Repository (File Component)

Stores and manages user conversations, mood tracking records, emotional wellness history, mental health assessments, intervention logs, and trusted mental health resources to enable continuous monitoring and personalized support.

Langflow workflow



IBM watsonx.ai

Granite Playground / Model settings

Save Info

Sessions +

Smart Farming Advice

Default Session

User

Which crop is suitable for black soil during winter season?

AI

granite-3-8b-instruct

For black soil during winter season, the most suitable crops are:

1. **Wheat** – Best suited for black cotton soil; requires cool climate and well-drained soil.
2. **Gram (Chickpea)** – Grows well in black soil and needs less water.
3. **Barley** – Suitable for winter and tolerant to dry conditions.
4. **Mustard** – Performs well in black soil with good drainage.
5. **Linseed (Flax)** – A good option for winter cultivation in black soil.

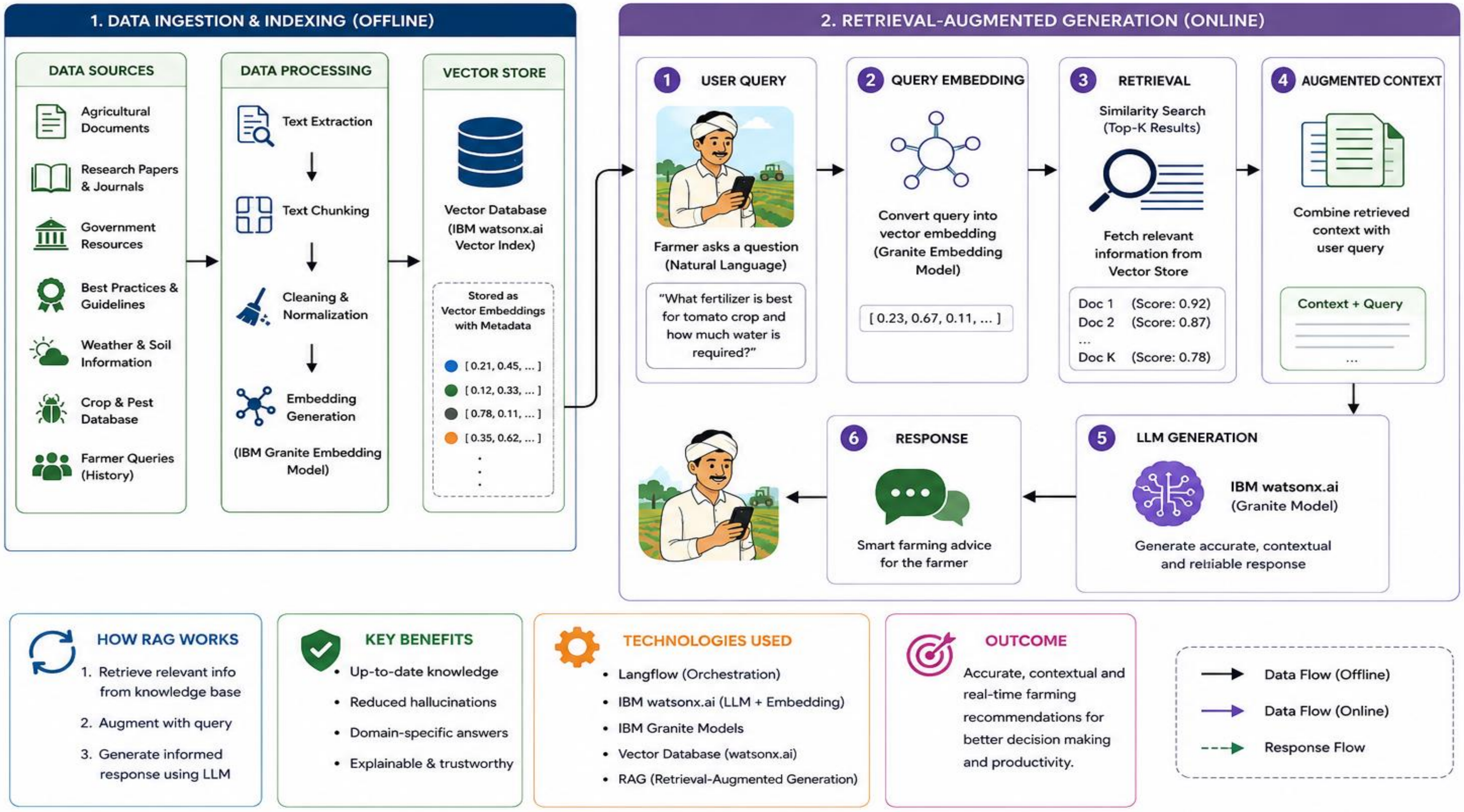
Farming Tips:

- Ensure proper irrigation but avoid waterlogging.
- Use organic manure or compost to improve soil fertility.
- Apply recommended fertilizers based on soil testing.
- Timely weeding and pest management are essential.
- Follow crop rotation to maintain soil health.

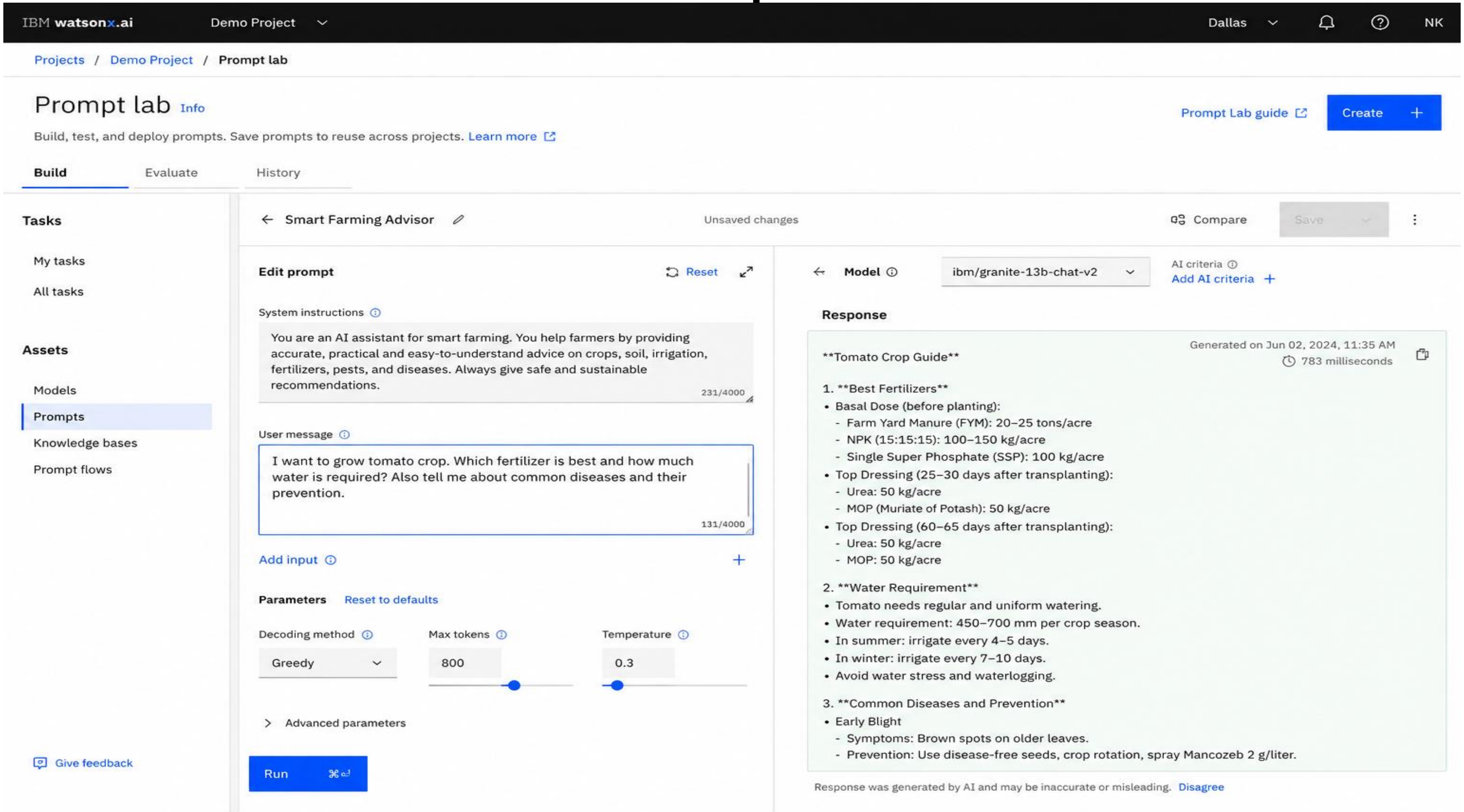
What is this document about?

RAG WORKFLOW

AI Agent for Smart Farming



IBM watsonx.ai Prompt Lab Screenshot



Role of Agentic AI in Smart Farming Advice

Agentic AI plays a crucial role in transforming traditional farming practices into intelligent, data-driven agricultural systems. Unlike conventional AI systems that only respond to predefined queries, Agentic AI can reason, make decisions, retrieve information, and coordinate multiple specialized agents to provide personalized farming recommendations.

In the proposed **AI Agent for Smart Farming Advice**, Agentic AI performs the following functions:

1. Intelligent Decision Making

Analyzes farmer queries and identifies the appropriate course of action.

Provides context-aware recommendations based on crop type, soil condition, weather, and farming practices.

2. Knowledge Retrieval using RAG

Retrieves relevant information from agricultural documents, government advisories, and farming guidelines.

Ensures that recommendations are accurate, up-to-date, and evidence-based.

3. Multi-Agent Collaboration

Agentic AI coordinates multiple specialized agents:

Crop Advisory Agent – Recommends suitable crops and cultivation practices.

Weather & Irrigation Agent – Provides weather-based irrigation guidance.

Pest & Disease Agent – Suggests preventive and corrective measures for crop diseases.

Market Insights Agent – Provides information on crop prices and market trends.

4. Personalized Recommendations

Generates customized advice according to farmer requirements, land conditions, crop selection, and environmental factors.

Helps farmers make informed decisions to improve productivity and profitability.

5. Real-Time Problem Solving

Responds instantly to farming-related queries.

Assists farmers in handling issues such as pest attacks, fertilizer usage, irrigation scheduling, and crop management.

6. Sustainable Farming Support

Encourages environmentally friendly farming practices.

Promotes efficient use of water, fertilizers, and other agricultural resources.

Technology Used

- **Langflow platform**

For designing and orchestrating multi-agent workflows visually

- **IBM Grainte model**

For eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization

- **IBM Cloud**

It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.

- **RAG**

RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.

- **IBM Watsonx.ai**

For model deployment, embeddings, and AI governance

Novelty and Uniqueness

The proposed **AI Agent for Smart Farming Advice** introduces an innovative approach to agricultural decision support by integrating **Agentic AI, Retrieval-Augmented Generation (RAG), Langflow, and IBM watsonx.ai** into a single intelligent platform. Unlike traditional farming advisory systems that provide static information, the proposed solution delivers personalized, real-time, and context-aware recommendations to farmers.

Key Novel Features

1. Multi-Agent Intelligent Architecture
2. Retrieval-Augmented Generation (RAG)
3. Personalized Farming Recommendations
4. Integration of Langflow and IBM watsonx.ai
5. Conversational Agricultural Assistant
6. Real-Time Decision Support

Uniqueness of the Proposed Work

The uniqueness of this project lies in its ability to combine:

Agentic AI
Multi-Agent Collaboration
RAG-based Knowledge Retrieval
IBM Granite Large Language Models
Langflow Visual Workflow Design
within a single smart farming platform.

Most existing farming applications focus on only one aspect such as weather forecasting or crop recommendation. In contrast, the proposed solution provides a **comprehensive AI-powered agricultural advisory ecosystem** that supports farmers throughout the crop lifecycle, from planning and cultivation to pest management and market decision-making.

Git Hub Link

https://github.com/Niky-Jain/NKJ_AI-Agent-Smart-Farming-Advice.git

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Future Scope of AI Agent for Smart Farming Advice

The proposed **AI Agent for Smart Farming Advice** has significant potential for future enhancement and large-scale deployment. With advancements in Artificial Intelligence, Internet of Things (IoT), cloud computing, and precision agriculture, the system can evolve into a comprehensive smart farming ecosystem.

1. Integration with IoT Sensors

The system can be integrated with IoT-based sensors to collect real-time farm data such as:

Soil moisture levels

Soil nutrient content

Temperature and humidity

Water availability

This will enable highly accurate and automated farming recommendations.

2. Real-Time Weather Forecasting

Future versions can integrate weather APIs to provide:

Rainfall predictions

Temperature forecasts

Drought alerts

Flood warnings

This will help farmers plan irrigation, sowing, and harvesting activities more effectively.

3. AI-Based Pest and Disease Detection

Using computer vision and deep learning techniques, farmers can upload crop images to:

Detect diseases at an early stage

Identify pests automatically

Receive instant treatment recommendations

Future Scope of AI Agent for Smart Farming Advice

4. Smart Irrigation Management

The AI agent can automatically control irrigation systems based on:

Soil moisture levels

Weather conditions

Crop water requirements

This will reduce water wastage and improve resource utilization.

5. Market Price Prediction

Future systems can analyze:

Historical market prices

Demand-supply trends

Seasonal variations

to predict crop prices and help farmers maximize profits.

6. Multilingual Voice-Based Assistant

The AI agent can support regional languages such as:

Gujarati

Hindi

Marathi

Tamil

allowing farmers to interact using voice commands and receive recommendations in their preferred language.

7. Drone and Satellite Integration

The system can be connected with drones and satellite imagery to:

Monitor crop health

Detect nutrient deficiencies

Identify pest infestations

Assess field conditions

on a large scale.

Future Scope of AI Agent for Smart Farming Advice

8. Personalized Farming Advisory

The AI agent can learn from:

Farmer profiles

Farm history

to provide highly customized recommendations.

9. Government and Agricultural Scheme Assistance

Future enhancements may include:

Information on government subsidies

Crop insurance guidance

to support farmers in accessing available benefits.

10. Smart Agriculture Ecosystem

The AI agent can be expanded into a complete agricultural platform connecting:

Farmers

Agricultural experts

Government agencies

Marketplaces

creating a digital ecosystem for efficient farm management.

11. Predictive Analytics for Crop Yield

Advanced machine learning models can forecast:

Expected crop yield

Disease risks

Water requirements

Fertilizer needs

helping farmers make proactive decisions.

12. Sustainable and Precision Farming

The system can promote:

Organic farming practices

Thank You!

Thank you for your time and interest.